Dynamics and Influences Analysis of Public Concerns in Mega Construction Projects: An Integrated Topic and Sentiment Modeling Approach

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Abstract: Mega construction projects (MCPs) are often subject to complex and evolving public concerns, with both thematic focus and sentiment attitudes evolving continuously throughout the project lifecycle. This study proposes an integrated topic and sentiment modeling approach to uncover the evolutionary patterns of public concerns and their emotional fluctuations. First, a continuous-time dynamic topic model (cDTM) is developed to extract and track key topics of public concern from unstructured textual data over time. Second, the performance of shallow learning, deep learning, and large language models is systematically compared for sentiment analysis, with the optimal model selected to quantify public sentiment at different phases. Third, a topic–sentiment coupling model is constructed to comprehensively evaluate the critical factors driving public attention and emotional trends, and propose targeted public opinion management strategies. Using a real-world MCP as a case study, the study analyzes 1,163 pieces of textual records and reveals clear phase-specific shifts in public concerns as the project progresses. While public sentiment is generally positive, a notable increase in negative emotions is observed during the construction, expansion, and operational transition phases. The proposed integrated modeling framework provides a scientific basis for public opinion management, stakeholder engagement, and the promotion of sustainable development in MCPs. **DOI: 10.1061/JCEMD4.COENG-17561**.

Introduction

Mega construction projects (MCPs), as a central driver of infrastructure development and economic growth, are typically characterized by substantial investment, long implementation cycles, and the involvement of multiple stakeholders (Ma and Fu 2020). In this context, the public is not merely a passive observer of the project but also a direct or indirect stakeholder. Their focal concerns and sentiment attitudes exert profound influence on the social acceptance, policy support, and long-term performance of MCPs (Xue et al. 2020). Public concerns differ across project phases: during the programming and design stages, attention is primarily directed toward project feasibility and functional design (Tu and Jin 2024; Wang et al. 2020), whereas in the construction and operational stages, safety risks and service quality become dominant issues of concern (Wu et al. 2019).

Failure to monitor and manage public concerns in a timely and effective manner may trigger a series of negative consequences. First, neglecting societal concerns can result in project approval delays, implementation setbacks, or even legal disputes and collective incidents, thereby amplifying uncertainty and cost risks (Flyvbjerg 2012). Second, the persistent spread of negative sentiments on social media can intensify social conflicts, damage the project's public image, and undermine governmental credibility (Yin et al. 2024). Third, the absence of dynamic feedback for public opinions weakens trust among stakeholders, potentially leading to social resistance or insufficient utilization during the operational phase, which ultimately diminishes the long-term value of projects (Wang and Li 2025).

In recent years, with the rapid proliferation of social media and digital platforms, discussions concerning MCPs have grown exponentially, generating vast volumes of unstructured textual data (Lobo and Abid 2020). Such data encapsulate the public's focal issues and emotional attitudes, providing researchers with unique opportunities to capture the dynamics of societal perceptions (Nik-Bakht and El-Diraby 2020). Among them, text mining methods, particularly topic modeling, can extract latent topics of public concern from large-scale corpora (Ezzeldin and El-Dakhakhni 2020); sentiment analysis techniques, especially large language models (LLMs), can capture complex emotional tendencies and semantic information, thereby characterizing the subtle fluctuations of public emotions (Min et al. 2024).

Nevertheless, existing studies often treat MCPs as static entities and lack systematic analyses of the dynamic evolution of public concerns throughout the entire project lifecycle (Xue et al. 2021). Moreover, topic modeling and sentiment analysis have seldom been effectively integrated, making it difficult to reveal the emotional mechanisms underlying public concerns and their implications for project management (Jin 2025). To address these gaps, this study proposes a dynamic framework that integrates topic modeling and sentiment analysis. By leveraging unstructured textual data from social media, news reports, and related platforms, the framework systematically uncovers the dynamic patterns of public concerns in MCPs and identifies their key influencing factors, thereby providing a scientific foundation for public opinion management across the full project lifecycle.

Research Background

Public Concerns in Mega Construction Projects

Early studies of public concerns primarily examined the impacts of the opinion on project implementation, focusing on issues such as ecological protection, resource allocation, and social equity

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(Biddulph 2011; Li et al. 2018). With the advancement of public participation theory, public concerns have increasingly been regarded as a key mechanism in stakeholder interaction. Their dynamic shifts not only shape the social acceptance of projects but also substantially determine the effectiveness of communication strategies, risk management, and decision-making optimization (Gao et al. 2022; Lobo and Abid 2020). Recent research has further highlighted the role of public engagement in enhancing transparency, strengthening social trust, and mitigating conflicts of interest, underscoring its significance in shaping the long-term social value of MCPs (Ural and Severcan 2022; Yu 2022; Zander et al. 2024).

Regarding the identification and assessment of public concerns, scholars have proposed various approaches to address their inherent complexity (Wang et al. 2020). Traditional methods, such as structured interviews and focus groups, reveal deep-seated public demands and emotional tendencies, offering rich qualitative insights (Sainati et al. 2020). Meanwhile, questionnaire surveys, as the quantitative method, can systematically capture the main concerns of the public and their influencing factors, with particular advantages in large-sample collection and trend analysis (Zander et al. 2024). However, these approaches often face challenges, including limited sample representativeness, high costs, and insufficient capacity to capture dynamics, making them inadequate for addressing the rapidly changing landscape of contemporary public opinion.

The digital era has greatly expanded research pathways. Time-series-based dynamic modeling has been widely applied to uncover the evolution of public concerns across project stages (Nokkaew et al. 2024). Recently, researchers have leveraged machine learning methods to advance the intelligence of public concern studies. Kundu et al. (2023) conducted an automated sentiment analysis of news reports on four major UK projects, offering a longitudinal comparison of public opinion dynamics across different project types. Nokkaew et al. (2023) applied deep learning and localized pretrained models to classify public comments on cross-border projects, demonstrating the value of language-specific models in enhancing analytical accuracy. Soomro et al. (2024) integrated sentiment analysis with spatiotemporal modeling in the case of Pakistan, showcasing the strategic value of social media data for crisis management and public engagement.

Notably, recent studies have introduced LLMs into the study of public participation in MCPs, significantly improving the capacity for large-scale text processing and semantic understanding. Wang et al. (2025) proposed an LLM-based framework for analyzing public participation and interaction networks in nine large projects in Hong Kong. This framework identified key opinion leaders and mapped dynamic patterns of public interactions, offering actionable insights for decision-makers in designing optimized communication strategies. In addition, Yu et al. (2025) expanded the application scope of surveys and interviews through modular artificial intelligence (AI) agents, enabling more interactive and transparent collection of public opinions, and demonstrating the potential of generative AI in transportation and infrastructure decision-making.

Overall, public concerns not only reflect societal attitudes and expectations toward projects but also constitute a fundamental basis for stakeholder collaboration and risk governance (Rodriguez-Melo and Mansouri 2011). Effective management of public concerns can optimize the social support environment of projects, mitigate misunderstandings and conflicts arising from information asymmetry, and enhance public satisfaction and project legitimacy (Wang et al. 2016).

Topic and Sentiment Modeling Approaches

In the context of large-scale textual data analysis, topic and sentiment modeling approaches provide powerful tools for identifying and assessing public concerns (Hemmatian and Sohrabi 2019). Topic modeling clusters key topics within texts, unveiling the core issues of public concerns (Luo and He 2021), while sentiment modeling analyzes the emotional attitudes expressed by the public, offering deeper insights into public stances and emotional fluctuations (Birjali et al. 2021). These methods have been widely applied across disciplines such as architecture, sociology, and computer science, serving as effective approaches for interpreting the dynamics of public concerns (Gao et al. 2020; Kumi et al. 2024; Zhong et al. 2024).

Topic modeling is a text mining technique based on probabilistic graphical models, aiming to extract latent topic structures from large volumes of unstructured text data (Yao et al. 2016). By constructing latent relationships between terms and documents, this technique enables the automatic identification of topic distributions within texts, providing a scientific method for recognizing and dynamically tracking public concerns (Koller and Friedman 2009). Among early topic modeling approaches, probabilistic latent semantic analysis (PLSA) assumes that documents are generated from a probability distribution over latent topics, forming a relational network between terms, documents, and topics (Hofmann 2001). Latent Dirichlet allocation (LDA), an extension of PLSA, introduces a Dirichlet prior distribution, assuming that each document is composed of a probabilistic mixture of multiple topics, and each topic is defined by a probability distribution over terms. This method effectively uncovers multitopic structures in texts and offers strong interpretability and scalability (Blei et al. 2003). Furthermore, to analyze the temporal evolution of topics, the continuous time dynamic topic model (cDTM) incorporates temporal information into the topic modeling framework, allowing for a more precise depiction of topic evolution over time (Wang et al. 2012). This model demonstrates significant advantages in exploring the long-term dynamics of public concerns and event-driven topic changes (Jähnichen et al. 2018).

Sentiment modeling aims to extract sentiment-related information from textual data to identify public emotional attitudes toward specific events or topics (Tang et al. 2017). It holds substantial value in understanding public opinion, predicting societal trends, and optimizing project communication strategies (Das and Singh 2023). Shallow learning methods, including support vector machines (SVMs), naïve Bayes (NB), and artificial neural networks (ANNs), achieve sentiment classification and regression analysis by designing feature extraction strategies (Vijayvergia and Kumar 2021). Deep learning techniques further enhance sentiment analysis capabilities—convolutional neural networks (CNNs) capture local sentiment features within texts, while recurrent neural networks (RNNs) and long short-term memory (LSTM) networks effectively model contextual sentiment information (Yang et al. 2021). In recent years, sentiment modeling based on LLMs has gained widespread attention. From Llama and Qwen to DeepSeek, these models leverage pretraining on vast corpora to learn rich linguistic features, significantly improving sentiment classification and sentiment tendency analysis (Min et al. 2024).

In summary, analyzing both the thematic content and the sentiment features of texts enables a more comprehensive understanding of public concerns (Wang et al. 2024). Moreover, topic–sentiment modeling approaches combined with dynamic time series analysis are rapidly evolving (Xue et al. 2021). Such methods offer significant advantages in analyzing the evolution of public concerns throughout the lifecycle of MCPs and assessing public emotional responses to different project decisions.

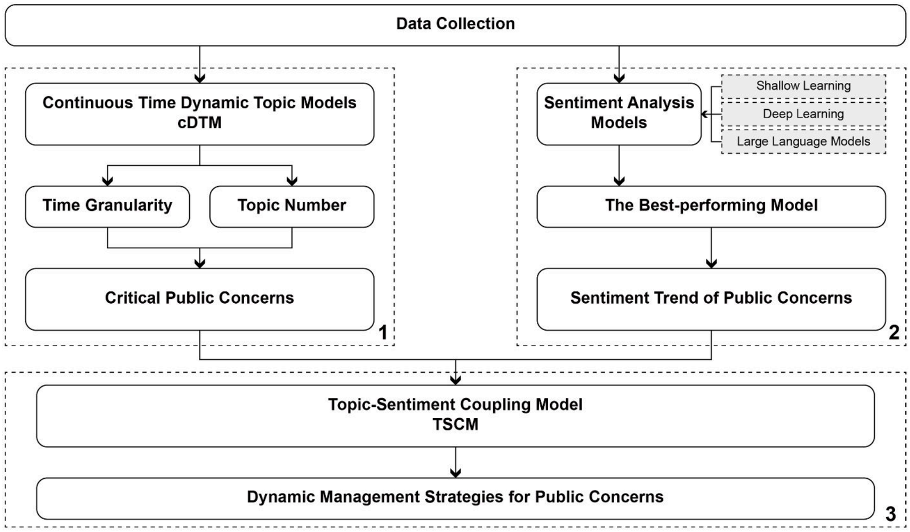

Fig. 1. Research framework of dynamic analysis on public concerns, showing Steps 1–3.

Research Methodology

As illustrated in Fig. 1, the research framework is divided into three steps. First, project-related textual data were collected from official sources, and a topic model was applied to identify public concerns, thereby revealing the focal issues of public attention across different project stages. Second, multiple sentiment analysis models were evaluated through comparative experiments, and the optimal model was selected to measure the sentiment orientation of public concerns, thus capturing shifts in public attitudes throughout the project lifecycle. Finally, by integrating the outcomes of topic identification and sentiment analysis, a topic–sentiment coupling model was constructed to assess the key drivers of public concerns at each stage and to propose targeted public opinion management strategies.

Continuous Time Dynamic Topic Model

The cDTM was developed as an extension of the traditional LDA, with its core concept introduced by Wang et al. (2012). Unlike the DTM, which relies on discrete time intervals, the cDTM adopts a continuous-time framework that directly incorporates timestamped textual data. This enables a more flexible and precise characterization of topic evolution across different time points. The graphical model of cDTM is illustrated in Fig. 2.

Specifically, each document d is assigned a timestamp t_d, and its topic distribution evolves. The distribution parameter $\beta_k(t)$ of each topic k changes continuously with time t, following the stochastic properties of Brownian motion. For any two time points t_i and t_j, the evolution of topic parameters is defined as

$$\beta_k(t_j) = \beta_k(t_i) + \epsilon_{ij}, \epsilon_{ij} \sim N(0, \sigma^2) \quad (1)$$

At timestamp t_d, the topic distribution of document d is denoted as $\theta_d(t_d)$. From this distribution, a topic k is sampled, which in turn generates word w_d^i according to the word distribution β_k. The corresponding probability equations are as follows:

$$P(z_d^i = k \,|\, \theta_d(t_d)) = \theta_d^k(t_d) \quad (2)$$

$$P(w_d^i \,|\, z_d^i = k, \beta_k) = \beta_k(w_d^i) \quad (3)$$

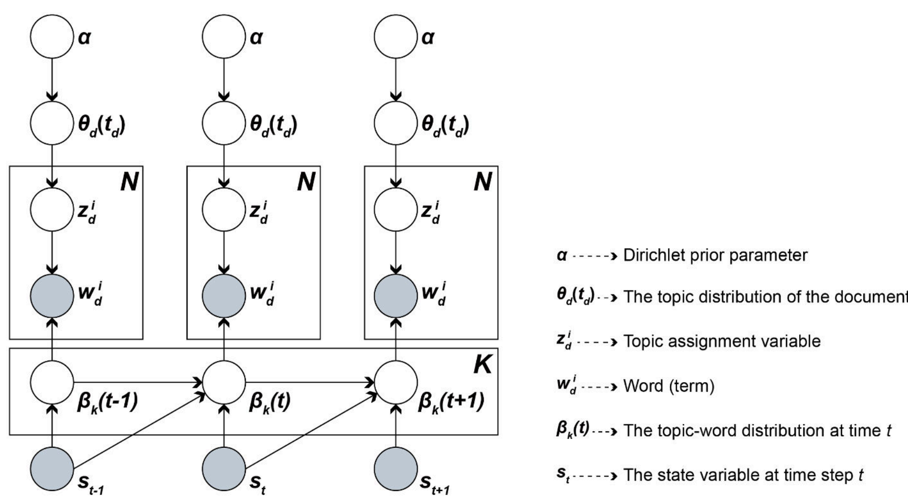

Fig. 2. Graphical model representation of the cDTM.

where $\theta_d^k(t_d)$ = probability that document d at time t_d belongs to topic k; and $\beta_k(w_d^i)$ = probability of generating word w_d^i under topic k. To approximate the posterior distribution, the variational Kalman filter algorithm is employed. For each document d, a variational distribution $q(\theta_d)$ is defined to approximate its true topic distribution. By maximizing the Kullback–Leibler divergence between the variational posterior and the true posterior, the optimal variational parameters can be obtained.

To systematically evaluate the performance of cDTM, this study introduced two metrics: per-word predictive perplexity and semantic coherence. The former tests the predictive capability of the model across different temporal granularities, defined as

$$Perplexity_{PW} = exp\left(-\frac{1}{N}\sum_{d=1}^{|D|}\sum_{w=1}^{|W_d|}\log P(w|d)\right) \tag{4}$$

where $P(w|d)$ = probability of predicting word w given document d; W_d = word set of document d; and $N = \sum_{d=1}^{|D|}|W_d|$ = total number of words in the corpus. A lower perplexity indicates stronger predictive capability and better temporal adaptability. Meanwhile, semantic coherence assesses the semantic relatedness of words within a topic by calculating the co-occurrence of high-probability terms under that topic, defined as

$$S(\theta_k) = \frac{1}{|V_k|}\sum_{i,j\in V_k}\log\frac{P(w_i, w_j|\theta_k)}{P(w_i|\theta_k)P(w_j|\theta_k)} \tag{5}$$

where V_k = set of high-probability words under topic k, w_i and w_j = words within this set; and $P(w_i, w_j|\theta_k)$ = their joint probability under topic k. Higher values indicate better semantic coherence within the topic.

Sentiment Analysis Model

This study employs three sentiment analysis methods, including shallow learning models, deep learning models, and LLMs, to assess the sentiment polarity (positive, neutral, negative) and intensity of public concerns (Jiang et al. 2016). Through comparative experiments, the objective is to determine the optimal sentiment analysis model, ensuring accuracy and reliability in subsequent research (Table 1).

Shallow learning models primarily rely on manually engineered features, performing well in scenarios with smaller data sets and lower feature dimensionality. They offer low computational costs and are suitable for fundamental sentiment analysis tasks. Deep learning models, in contrast, learn contextual relationships in text and capture long-range dependencies, making them more effective for complex sentiment classification in large-scale textual data. LLMs, pretrained on extensive corpora, exhibit superior semantic comprehension, enabling them to recognize subtle sentiment variations, particularly in tasks requiring deep contextual understanding.

To ensure a fair comparison across different models, this study optimizes each model's hyperparameters (Table 2). Shallow learning models are fine-tuned by adjusting regularization parameters, loss functions, and iteration counts to enhance classification performance. Deep learning models are standardized with consistent vocabulary size, maximum sentence length, batch size, and learning rate to maintain comparability. LLMs are optimized on the Ollama platform to ensure consistency in computational environment and data processing (Marcondes et al. 2025). All models are trained on the same data set and evaluated on the test set. Performance is measured using four quantitative metrics, including accuracy, recall, precision, and F1-score, to assess sentiment classification and intensity prediction capabilities.

Following sentiment analysis, this study further calculates the public perception value to quantify the overall sentiment trend of public concerns toward MCPs. This metric integrates sentiment

Table 1. Performance evaluation of sentiment analysis models

Model type	Models
Shallow learning	Naive Bayes (NB), support vector machine (SVM), logistic regression (LR), artificial neural network (ANN)
Deep learning	Long short-term memory (LSTM), bidirectional LSTM (BiLSTM), attention-based BiLSTM, BERT
LLMs	Qwen, Llama, DeepSeek

Table 2. Parameters of sentiment analysis models

Model type	Model	Parameters	Value
Shallow learning	NB	Alpha	1.0
		Fit prior	True
	SVM	Penalty	L2
		Loss	Hinge
		Maximum iterations	1,000
		Learning rate	0.01
	LR	Penalty	L2
		Solver	liblinear
		Maximum iterations	1,000
	ANN	Activation	ReLU
		Solver	Adam
		Maximum iterations	200
		Learning rate	0.001
Deep learning	LSTM, BiLSTM, BiLSTM + Attention	Vocabulary sizes	10,000
		Class number	3
		Maximum sentence length	128
		Embedding dimension	300
		Batch size	32
		Epochs	48
		Learning rate	0.001
		Layers	2
	BERT	Pretrained model	BERT-base-uncased
		Maximum sentence length	128
		Batch size	16
		Epochs	24
		Learning rate	2e-5
		Optimizer	AdamW
LLMs	Qwen2.5	Pretrained model	7B
	Llama3.1	Pretrained model	8B
	Deepseek-R1	Pretrained model	7B

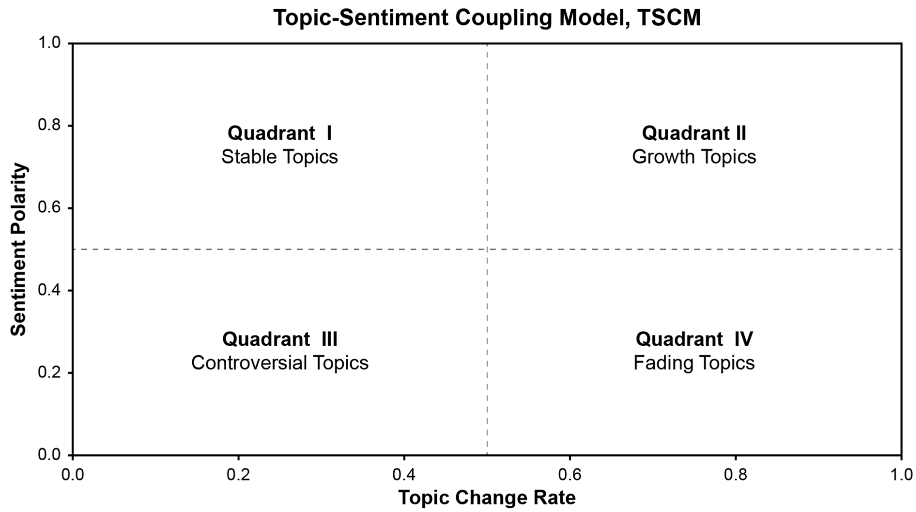

Fig. 3. Topic–sentiment coupling model for managing public concerns.

polarity, public attention, and topic similarity. It is computed using latent semantic analysis (LSA) combined with K-means clustering (Yan et al. 2017). The equation is as follows:

$$Value_{Public\ Perception} = \frac{1}{N} \cdot \sum_{A=1}^{N} \left(\frac{R_A \cdot S_A}{N_A \cdot N_A} \right) \tag{6}$$

where N = total number of clusters; N_A = number of texts within cluster A; R_A = total sentiment score of texts in cluster A (sentiment polarity is converted to numerical values: positive = 10, neutral = 5, negative = 1); and S_A = total topic similarity within cluster A.

Topic–Sentiment Coupling Model

To capture the evolution of public concern topics and their corresponding emotional dynamics, this study proposes a topic–sentiment coupling model (TSCM), designed to systematically analyze the coupling relationship between the topic change rate (TCR) and sentiment polarity (SP) over time. Previous research has attempted to integrate topic modeling with sentiment analysis. The topic–sentiment mixture (TSM) model (Mei et al. 2007) and the joint sentiment–topic (JST) model (Lin and He 2009) explored joint modeling of topics and sentiments. Subsequent studies extended these frameworks, developing approaches such as the JST model of reviews and ratings (JST-RR) and the labeled joint sentiment topic (LJST) model to analyze public opinion dynamics (Liang et al. 2023; Sengupta et al. 2021). However, these approaches primarily relied on discrete-time modeling and did not explicitly account for the role of TCR in managing public concerns. In contrast, TSCM adopts a continuous-time framework, introducing TCR and SP as key indicators and employing strategic quadrant analysis to identify distinct patterns of topic evolution.

Specifically, TCR measures the dynamic evolution of topics within public concerns. It is calculated based on the topic distribution parameters $\beta_k(t)$ extracted by cDTM, combined with term frequency-inverse document frequency (TF–IDF) weights, thereby capturing fluctuations in topical attention more accurately

$$TCR_i = \frac{\sum_{t=1}^{n-1} |w_i^{(t+1)} - w_i^{(t)}|}{n-1} \tag{7}$$

where $w_i^{(t)}$ = weight of topic i at time t; and n = total number of time segments. A higher TCR indicates larger fluctuations across time, typically corresponding to short-term hotspots or sudden events, whereas lower TCR values often signal long-term, stable topics of concern. SP, by contrast, measures the overall sentiment orientation of the public toward a specific topic, taking into account both sentiment scores and text frequency

$$SP_i = \frac{\sum_{t=1}^{n} s_i^{(t)} \cdot f_i^{(t)}}{\sum_{t=1}^{n} f_i^{(t)}} \tag{8}$$

where $s_i^{(t)}$ = average sentiment score of topic i at time t (ranging from –1 to 1); and $f_i^{(t)}$ = total frequency of documents associated with the topic at time t. This ensures that high-frequency topics contribute more substantially to the overall sentiment polarity. To mitigate the impact of short-term fluctuations, this study uses sliding window smoothing and normalizes TCR and SP

$$TCR_i^{norm} = \frac{TCR_i - \min(TCR)}{\max(TCR) - \min(TCR)} \tag{9}$$

$$SP_i^{norm} = \frac{SP_i - \min(SP)}{\max(SP) - \min(SP)} \tag{10}$$

where the normalized values TCR_i^{norm} and SP_i^{norm} are restricted to the range [0,1], thereby ensuring comparability across different topics. Finally, based on the computed TCR and SP values, a strategic quadrant map of public concerns (Fig. 3) is constructed. Topics are categorized into four types according to their position in the two-dimensional space: growth-oriented topics (high TCR, positive SP): rapidly increasing in attention with positive public sentiment, typically representing emergent hotspot issues; declining topics (high TCR, negative SP): rapidly increasing in attention but associated with negative sentiment, often reflecting sudden controversies or crises; stable topics (low TCR, positive SP): sustained attention with positive public sentiment, generally forming core consensus issues; controversial topics (low TCR, negative SP): steady attention but negative sentiment, often pointing to persistent social conflicts or long-standing disputes.

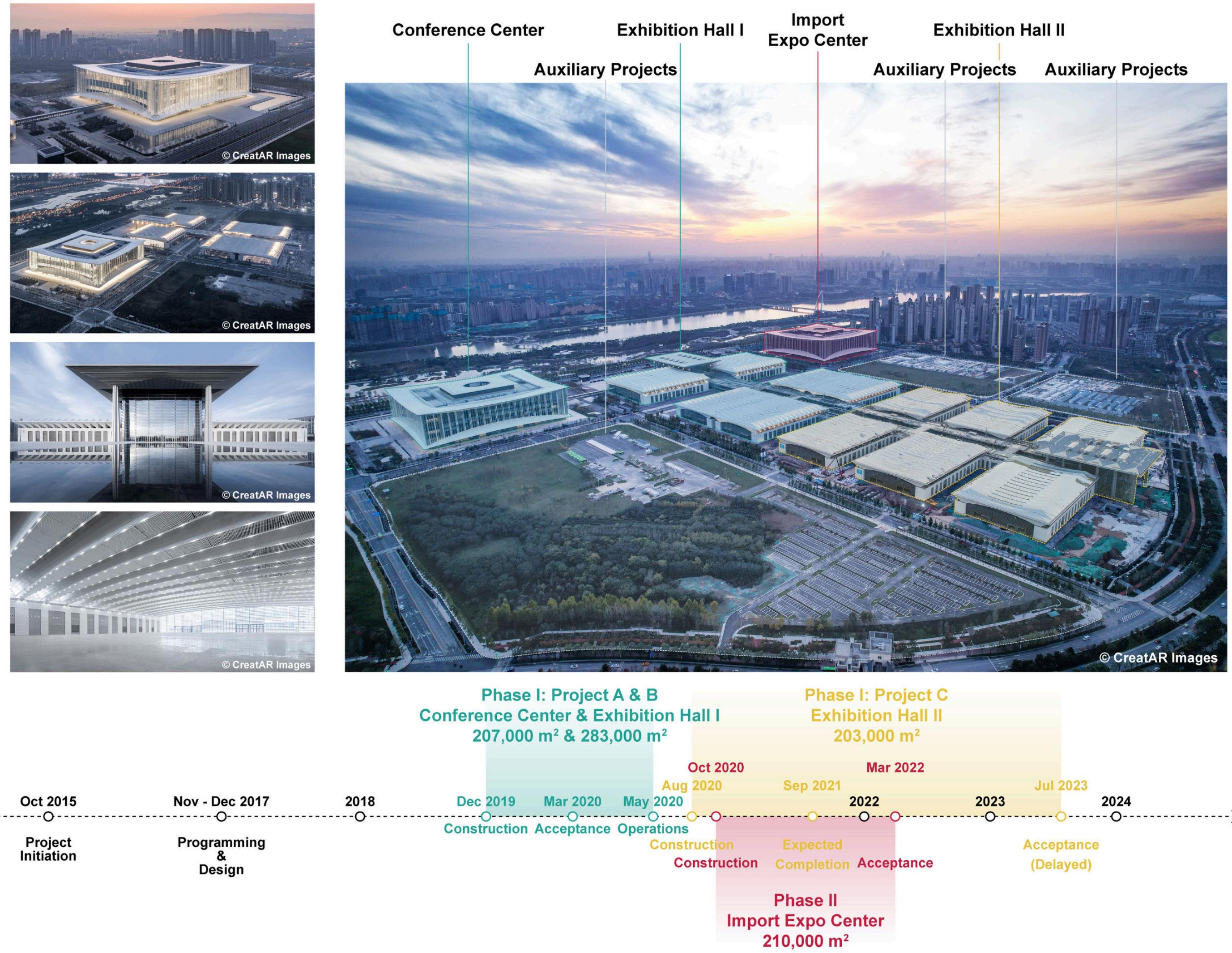

Fig. 4. XIANICEC project and its time phase. (Images courtesy of CreatAR Images, © CreatAR Images.)

Case Study

Case Selection

This study selects the Xi'an International Convention and Exhibition Center (XIANICEC) project as the case study. As the permanent venue for the Belt and Road Forum for International Cooperation, XIANICEC holds significant political, economic, and cultural importance. Covering a total area of approximately 1,200 mu, the project involves a massive investment and multiple construction phases, making it a typical MCP in China (GMP 2020; TJAD 2020). Given its scale and complexity, the project is not only crucial for the economic and urban development of Xi'an and western China but also plays a key role in enhancing international influence and driving regional economic growth.

Since its planning and approval in 2015, the XIANICEC project has undergone multiple construction phases (Fig. 4). Architectural design commenced in 2017, followed by construction in 2019, which was implemented in two phases. The first phase was completed and put into operation in 2020, while the second phase was completed in 2022 but has not yet been fully operational. Currently, supporting residential, commercial, and public infrastructure is being developed in an orderly manner. This long-term and complex construction process provides an abundance of public concern data, allowing this study to comprehensively examine the evolution of public sentiment across different MCP phases and its potential impact.

Data Collection and Preprocessing

The public concern data in this study are sourced from official XIANICEC platforms, including Weibo, WeChat public accounts, news reports, and government documents. These sources provide a comprehensive reflection of public discussions across various project phases, covering topics from project initiation, programming, and design to construction and operation. Social media platforms, as primary channels for public opinion expression, offer high-frequency and real-time feedback, revealing the diversity and timeliness of public sentiment (Bertot et al. 2012). In contrast, news reports and government documents provide authoritative and formal public discourse, helping to mitigate information noise from social media and enhance the reliability of the analysis results (Xue et al. 2020). By integrating data from these sources, this study aims to capture the dynamics of public concerns and sentiment evolution comprehensively, ensuring the representativeness of the result.

During the data collection process, a total of 1,718 public-concern-related textual records were retrieved from multiple

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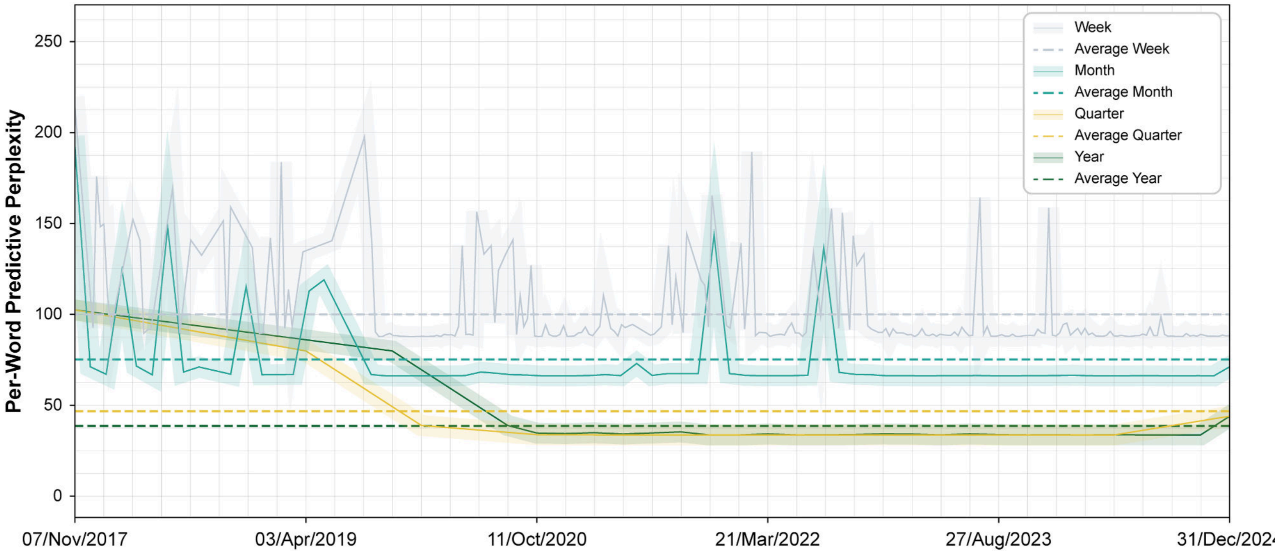

Fig. 5. Comparison results of per-word predicted perplexity.

platforms using web crawlers. To ensure compatibility with the analytical models, a systematic data preprocessing procedure was conducted. First, stop word filtering was applied to remove common but semantically insignificant high-frequency words, such as "de" (of), "shi" (is), and "zai" (in). Second, word segmentation was performed using the Jieba Chinese word segmentation tool, breaking sentences into meaningful words or phrases. Additionally, to ensure accurate processing of domain-specific terminology, a custom lexicon for the construction management domain was developed, covering terms such as "construction safety" and "environmental pollution." Third, noise reduction was conducted by filtering out advertisements, meaningless comments, and other irrelevant content to enhance the purity and quality of the data set. Following these preprocessing steps, a final data set of 1,163 structured, high-quality textual records was obtained, providing a robust data foundation for subsequent topic modeling and sentiment analysis.

In terms of sentiment annotation, 30% of the data set was randomly selected for manual labeling to train and evaluate the sentiment analysis model. The annotation was independently conducted by two experts with backgrounds in architecture and natural language processing. Sentiment categories were divided into three classes: positive, neutral, and negative. Texts containing positive emotional words or phrases (e.g., "excellent," "convenient transportation") were labeled as positive; those containing negative expressions (e.g., "poor," "inadequate facilities") were labeled as negative; while texts that only stated facts or showed no clear emotional tendency were labeled as neutral (Appendix SI). To ensure annotation quality, both experts first labeled the data set independently and subsequently discussed and reconciled samples with discrepancies, thereby producing a standardized sentiment corpus. The interannotator agreement was assessed using Cohen's Kappa coefficient, which yielded a score of 0.82, indicating a high degree of consistency between annotators in assigning sentiment labels. Finally, the manually annotated data set was divided into a training set (70%), test set (20%), and validation set (10%).

Topic Model Inference and Optimization

To determine the optimal time granularity for the cDTM, this study segmented the data using four different time units: weekly, monthly, quarterly, and annually. The per-word perplexity was calculated for each time granularity to evaluate its impact on topic modeling. As shown in Fig. 5, the data set segmented annually exhibited the lowest perplexity, indicating the highest topic stability at this granularity. This suggests that an annual segmentation effectively reduces noise interference and accurately captures the evolution of topics. In comparison, quarterly granularity yielded results close to those of the annual segmentation but still exhibited some degree of instability. Monthly and weekly granularities resulted in higher perplexities, indicating that an overly fine-grained time division fragments the data, thereby compromising topic coherence. Considering both topic stability and the rationality of time segmentation, this study ultimately selected annual granularity as the time unit to ensure the continuity and interpretability of the topic evolution analysis.

After determining the optimal time granularity, this study further utilized the topic coherence score to identify the optimal number of topics for each time period. An iterative simulation was conducted with topic numbers ranging from 1 to 30, and the coherence score was computed for different years. As shown in Appendix SII, public discussions between 2017 and 2019 were relatively focused, leading to a lower number of optimal topics. In 2020–2022, public concerns became increasingly diverse, reflecting a broader discussion scope. By 2022–2024, the range of public discussions further expanded, reaching the peak number of optimal topics, indicating a notable increase in both the breadth and depth of public concerns in the XIANICEC project. Based on this analysis, this study determined the optimal number of topics for each year and constructed the optimal cDTM for this case (Appendix SIII).

Sentiment Model Testing and Evaluation

Before performance testing, four deep learning models were trained according to the settings listed in Table 2. The accuracy and loss curves of the training process are shown in Fig. 6. As the number of training epochs increased, the accuracy of all models steadily improved and plateaued between the 15th and 20th epochs. Meanwhile, the loss curves gradually declined and stabilized within the same range, indicating that the models had converged. To ensure training stability and mitigate overfitting, the number of epochs was fixed at 20 for the final training configuration.

To comprehensively compare the performance of shallow learning models, deep learning models, and LLMs, four evaluation metrics were employed, as summarized in Table 3 (environmental

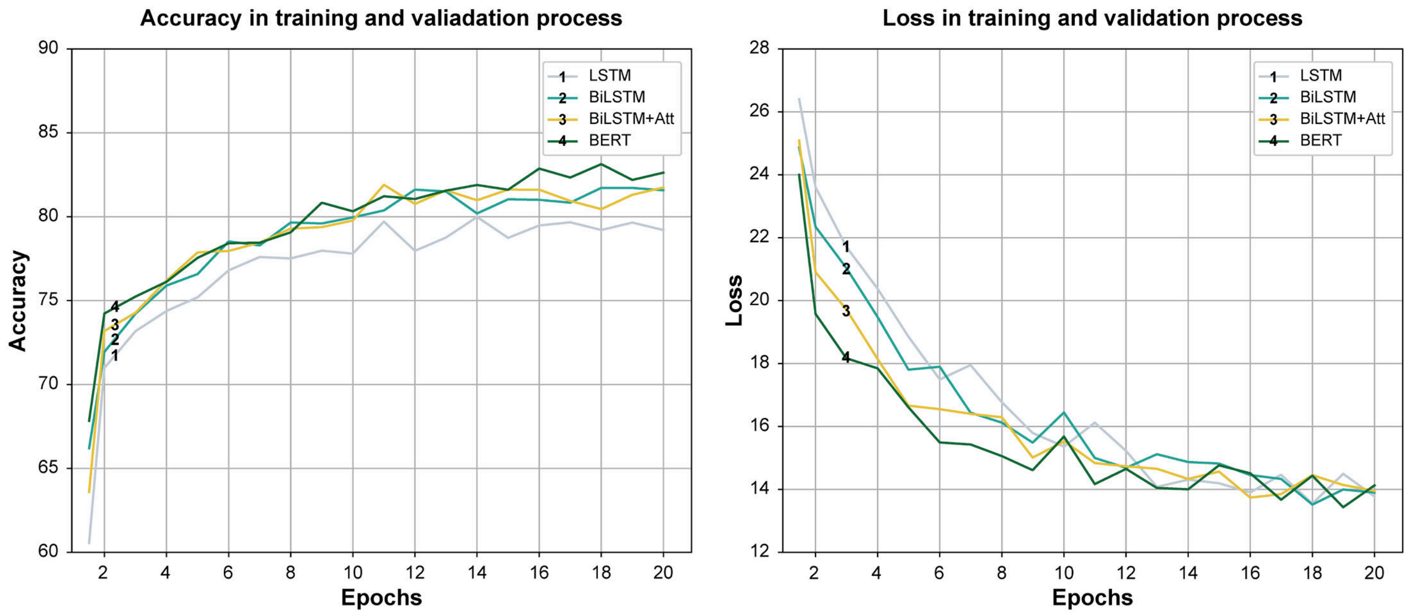

Fig. 6. Accuracy and loss rate curves of four deep learning models.

Table 3. Experimental results of each sentiment analysis model

Model type	Model	Accuracy	Precision	Recall	F1 score
Shallow learning	NB	0.789	0.758	0.770	0.764
	LR	0.799	0.769	0.776	0.772
	SVM	0.805	0.786	0.792	0.789
	ANN	0.811	0.792	0.784	0.788
Deep learning	LSTM	0.865	0.838	0.849	0.843
	BiLSTM	0.889	0.858	0.864	0.861
	BiLSTM + Att	0.900	0.874	0.880	0.877
	BERT	0.912	0.890	0.895	0.892
LLMs	Qwen2.5	0.939	0.926	0.928	0.927
	Llama3	0.935	0.922	0.925	0.923
	DeepSeek-R1	0.936	0.920	0.924	0.922

settings detailed in Appendix SIV). The shallow learning models demonstrated relatively stable performance, among which the ANN achieved the best accuracy at 81.1%. The deep learning models outperformed shallow ones overall, with the bidirectional encoder representations from transformers (BERT) model achieving the highest accuracy of 91.2%. LLMs delivered the best performance across all models, with accuracy consistently exceeding 93%, attributable to their superior ability to capture sentiment-related semantic relationships and contextual dependencies within text. Among them, the Qwen2.5 model achieved an accuracy of 93.9%, significantly outperforming the others.

Based on these testing and evaluation results, Qwen2.5 was selected as the optimal sentiment analysis model for subsequent dynamics analysis of public concern.

Research Findings

Key Topics of Public Concerns

Based on the optimized cDTM, this study extracted the topics of public concerns at different stages of the XIANICEC project and visualized their relationships through an intertopic distance map constructed with multidimensional scaling (Fig. 7). The results reveal that the high-frequency thematic terms of public concerns exhibited significant stage-specific variations as the project advanced.

During the programming stage, public discussions were concentrated on overall planning, investment scale, and functional positioning. The terms "international" and "comprehensive" reflected expectations for internationalization and integrated functions, while "budget" and "scale" highlighted concerns about economic feasibility. Meanwhile, "concert" and "investment promotion" indicated anticipation of future commercial and cultural value.

As the project entered the construction stage, public concerns gradually shifted toward construction processes and their impacts. In 2019, the keywords "traffic," "parking," and "traffic flow" illustrated concerns regarding traffic order and environmental pressure, whereas "equipment," "noise," and "progress" emphasized issues of construction quality, pollution, and schedule delays. By 2020, discussions extended to the architectural form: "modeling" suggested attention to building aesthetics, while "noise" continued to signify environmental disturbance.

With the commissioning of Sections A and B, public concerns shifted toward operations and user experience. In 2021, the terms "venue,","parking," and "entrance" highlighted priorities regarding convenience and accessibility, while "exhibition" indicated that operational activities had become a central issue. In 2022, the frequent appearance of "service," "subway," and "greening" demonstrated increasing public attention to transport integration, environmental improvement, and service quality, while "reservation" signaled that management systems had entered the public discourse.

By 2023, with the completion of Section C, evaluations of service quality and space utilization became more positive. Terms such as "good," "large," and "scenery" conveyed overall satisfaction, though "parking lot" continued to underscore the importance of supporting facilities. In 2024, the focus of discussions further converged on exhibition experience, industrial development, and functional optimization. Notably, "interaction" and "lakes bay" revealed emerging topics of intelligent experiences and the nighttime economy.

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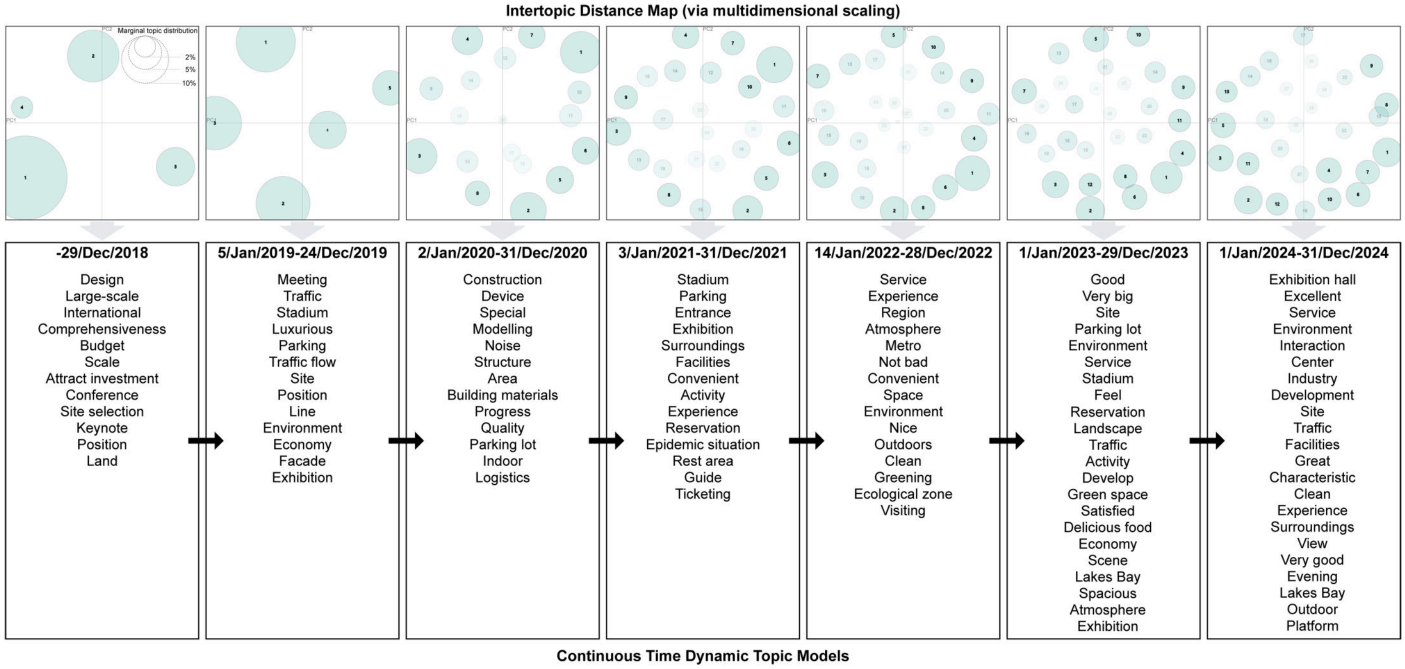

Fig. 7. Modeling results of the cDTM.

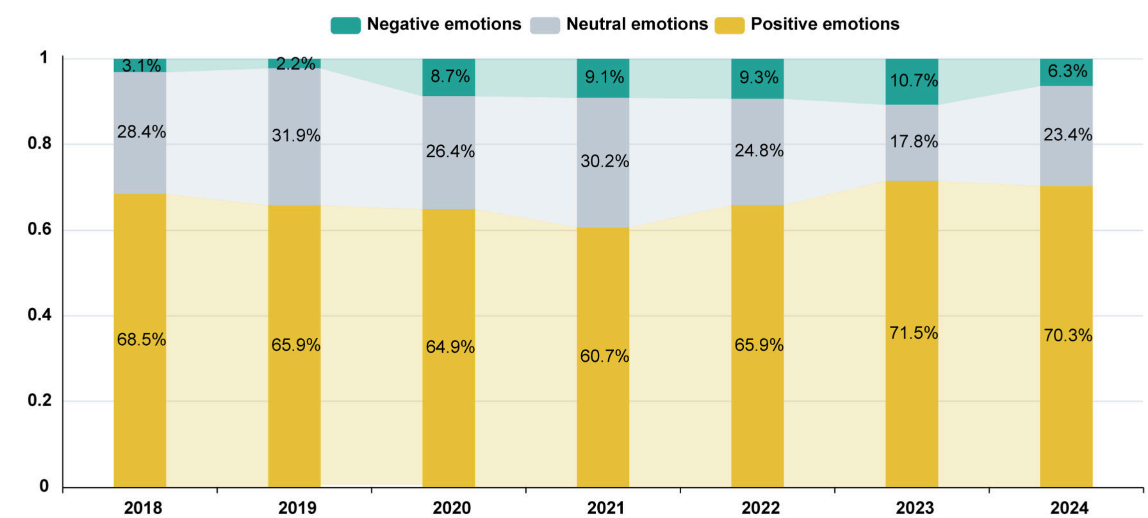

Fig. 8. Dynamic sentiment of public concerns.

Dynamic Sentiment of Public Concerns

To reveal the annual variation in public sentiment toward the XIANICEC project, this study employed the Qwen pretrained model to analyze sentiment polarity (Fig. 8). Overall, public attitudes were predominantly positive, with the proportion of positive sentiment exceeding 60% in each year. The lowest level was observed in 2021 (60.7%), primarily due to deficiencies in early-stage operations, while the peak occurred in 2023 (71.5%), driven by the maturity of venue operations, the abundance of exhibition activities, and the improvement of the industrial chain. Negative sentiment accounted for a small share in 2018–2019 (3.1% and 2.2%, respectively), but increased steadily thereafter, reaching its maximum in 2023 (10.7%). This trend indicates that public attention was limited in the early phase, whereas with construction progress and operational adjustments, concerns over traffic congestion, environmental impacts, and management issues became the major sources of negative discourse.

To further quantify public concern trends, this study calculated the perception value, as presented in Table 4. During 2018–2019, the perception value remained relatively low (2.889–2.930), suggesting that discussions were limited in the programming and design stages, and overall attitudes were relatively neutral. Between 2020 and 2021, the perception value rose sharply (5.305–5.425), indicating that construction-related issues such as progress, traffic, and noise stimulated extensive public discussion, rapidly elevating the level of perception. In 2022–2023, the perception value showed slight fluctuations (5.378–5.252). On one hand, the increased frequency of venue usage and the growing number of activities pushed the share of positive sentiment to its peak in 2023; on the other hand, expansion works and operational adjustments triggered skepticism and debate, resulting in the highest proportion of negative sentiment

Table 4. Results of sentiment ratio and perceived value

Year	Negative (%)	Neutral (%)	Positive (%)	Perception value
2018	3.1	28.4	68.5	2.889
2019	2.2	31.9	65.9	2.930
2020	8.7	26.4	64.9	5.305
2021	9.1	30.2	60.7	5.425
2022	9.3	24.8	65.9	5.378
2023	10.7	17.8	71.5	5.252
2024	6.3	23.4	70.3	5.767

during this period. By 2024, the perception value climbed to its highest level (5.767), indicating that as venue operations stabilized and supporting infrastructure became more complete, overall public recognition and satisfaction with the project reached a new peak.

Strategic Mapping of Public Concerns

Based on the TSCM, this study constructed a strategic quadrant map of public concerns for the XIANICEC project (Fig. 9), illustrating the coupling relationship between public concern topics and sentiment across different development stages. The map is divided into four phases: programming and design, construction and development, operation and expansion, and maturity and optimization.

During the programming and design phase, the positioning of the international convention center generated widespread discussion. Among the emerging topics, “conference” and “international” reflected the public’s expectations that the project would enhance the city’s international influence and stimulate the growth of the high-end convention and exhibition industry. By contrast, “budget” exhibited a declining trend, as early discussions surrounding investment scale and fiscal pressures carried predominantly negative sentiment, focusing on issues such as funding sources and cost control.

As the project entered the construction phase, public attention shifted toward engineering quality, construction progress, and environmental impacts. “Progress” emerged as a growth-oriented topic, with the public closely monitoring whether the project could be completed on schedule amid challenges posed by pandemic controls and construction complexities; overall sentiment was relatively positive. Meanwhile, “construction” and “equipment” remained stable topics, reflecting sustained concern for safety, construction standards, and technical specifications.

Once the project was put into use, public discussion increasingly emphasized operational performance, user experience, and supporting facilities. “Experience” and “exhibition” became growth-oriented topics, as the diversity and interactivity of exhibition activities were widely recognized, leading to a notable increase in positive sentiment. However, “parking lot” showed a declining trend, with insufficient capacity, unreasonable fees, and peak-hour congestion triggering considerable negative evaluations.

By 2024, as the project entered its maturity phase, public concerns gradually extended to long-term issues such as industrial development, economic benefits, and urban influence. “Surroundings” emerged as a growth-oriented topic, reflecting public expectations regarding the industrial chain effects of the convention economy, business opportunities generated by international exhibitions, and the improvement of the surrounding commercial and cultural ecosystem.

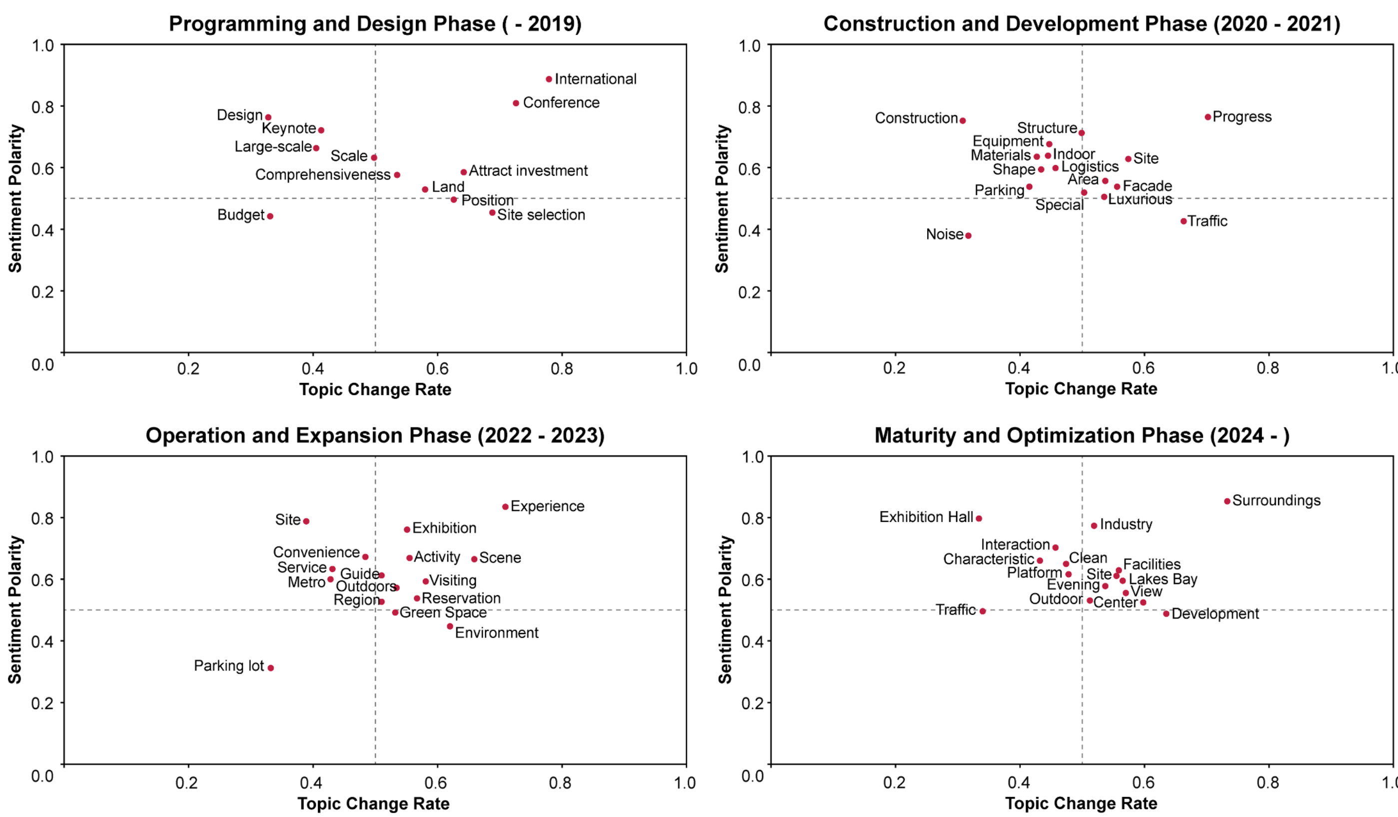

Fig. 9. Strategic quadrants of public concerns of the XIANICEC project.

Discussion

Key Drivers of Public Concerns

Across the full lifecycle of MCPs, the topics of public concerns and their associated sentiment trajectories are not random fluctuations but are shaped by multiple driving factors. First, changes in policies and regulations exert a direct influence. Both national-level development strategies and adjustments to local administrative regulations can significantly alter public sensitivity toward certain issues in the short term. Second, media outlets and social platforms serve as critical channels for shaping and amplifying public concerns. While media reports construct the public's initial perception of projects, social platforms—with their rapid information diffusion and participatory diversity—facilitate the swift spread of opinions and the polarization of sentiments. Third, unexpected events often act as turning points in the dynamics of public concerns. Incidents such as construction accidents or broader social crises can not only shift the focal points of public attention but also intensify emotional responses, thereby redirecting the trajectory of public discourse. These key drivers interact with one another, collectively shaping the evolution of public concerns at different stages of MCPs.

Stage-Specific Characteristics of Sentiment Evolution

The evolution of public sentiment toward MCPs exhibits both stage-specific and heterogeneous features. In the case of the XIANICEC project, public sentiment followed a progression from expectation to anxiety, and eventually to adaptation. During the programming and design stages, positive emotions dominated, reflecting optimism regarding the project's potential economic and social value, though concerns about feasibility and financial arrangements persisted. Once the project entered the construction stage, negative emotions increased markedly, driven by the immediate impacts of noise, traffic disruptions, and safety risks. Similar emotional fluctuations have been observed in international MCPs in the UK, South Korea, and Indonesia (Kundu et al. 2023; Lee et al. 2017). As the project transitioned into the operational stage, public sentiment gradually shifted toward positivity; however, recurring issues such as service quality deficiencies in management continued to trigger criticism. This coexistence of "recognition and skepticism" has also been documented in other operational stages of MCPs, such as London's Crossrail project and Tokyo's Olympic project, suggesting that the tension between symbolic value and functional value is a widespread phenomenon (Lobo and Abid 2020).

Practice-Oriented Strategies for Public Concern Management

Based on the findings of public concerns and sentiment evolution, stage-specific management strategies can be proposed. In the programming stage, enhancing information transparency serves as the foundation for building public trust. Governments and project owners should adopt phased information disclosure to release key data on budgets, approvals, and feasibility assessments, thereby reducing uncertainties and mitigating public skepticism (Bello et al. 2024). In the design stage, the public's sensitivity to environmental sustainability is increasingly pronounced. Hence, low-carbon design, energy-saving technologies, and ecological compensation measures should be incorporated into the project evaluation system, while public open discussions should be established to strengthen social oversight. During the construction stage, project managers need to implement intelligent construction information and real-time risk warning systems to minimize negative impacts on surrounding residents and commuting populations, thereby preventing the prolonged accumulation of negative emotions. In the operation stage, the long-term vitality of a project depends on the sustained creation of social value. Accordingly, it is essential to introduce public feedback into operation management, integrating service quality, facility maintenance, and social inclusiveness into performance evaluation criteria.

Limitations of the Modeling Approach

Although the integrated topic–sentiment modeling approach proposed in this study demonstrates good performance in revealing the evolutionary patterns of public concerns, it is not without limitations and uncertainties. First, cDTM is sensitive to time granularity and hyperparameter settings. Excessively fine or coarse time segmentation may lead to topic drift, thereby undermining the stability of the model. While this study optimized model outputs through perplexity and coherence metrics, the resulting topic structures may still be influenced by uneven data distributions and parameter configurations. Second, sentiment analysis entails inherent uncertainties across different linguistic contexts. In particular, sarcasm, metaphor, and semantically ambiguous expressions may lead to misclassification, as both shallow/deep learning models and LLMs may struggle under complex discourse conditions. Such errors can be further amplified when coupled with topic modeling, introducing additional uncertainty into the results. Future research could mitigate these issues by incorporating uncertainty interval reporting, fine-tuning LLMs, and leveraging multimodal data sources, thereby enhancing the robustness of the proposed approach.

Conclusions

This study proposes a dynamic analysis approach for public concerns based on integrated topic and sentiment modeling, using the XIANICEC project as a case study to uncover the evolutionary patterns of public concerns and their sentiment fluctuations throughout the lifecycle of MCPs. The findings indicate that public sentiment was predominantly positive overall, but exhibited certain negative fluctuations during the construction and expansion/operational adjustment phases, primarily due to construction disturbances, management changes, and uncertainties. As the project entered a stable operational phase, public concerns gradually shifted toward facility management, service experience, and social value, with sentiment trending more positively. These dynamics not only reveal the core concerns of the public at different stages but also reflect the broad impacts of MCPs on the environment, transportation, governance, and socio-cultural dimensions.

The contributions of this study are twofold. First, this study advances an integrated modeling framework that combines text mining, machine learning, and topic–sentiment coupling, thereby enabling the quantitative measurement of public concerns and sentiment trends, providing a novel technical pathway for intelligent public opinion management. Second, this study identifies the dynamic evolutionary patterns of public concerns in MCPs, revealing the mechanisms of concern migration and sentiment fluctuation, and thus offering theoretical insights and empirical evidence to support social recognition and governance optimization in MCPs.

However, this study has certain limitations. First, the analysis focuses solely on one project in China. While representative, the generalizability of the findings across projects and regional contexts remains to be further validated. Second, the data were primarily collected from official channels and dominant social platforms. This limited diversity in data sources may have overlooked public voices from decentralized or niche platforms, introducing potential bias. Future research could expand by conducting cross-project and cross-regional comparative analyses to test the applicability of the proposed approach under different socio-cultural contexts. Additionally, broadening the sources of public opinion data to include more diverse online communities and platforms would improve the generalizability of the conclusions.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the author upon reasonable request.

Author Contributions

Shitao Jin: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Project administration; Resources; Software; Supervision; Validation; Visualization; Writing – original draft; Writing – review and editing.

Supplemental Materials

Appendixes SI–SVII are available online in the ASCE Library (www.ascelibrary.org).

References

Bello, A., A. A. Abdulraheem, O. P. Afolabi, A. Aka, and P. O. Gbenga. 2024. "Assessing the underlying factors affecting trust and transparency in the construction industry: A mixed method approach." *Constr. Econ. Build.* 24 (1–2): 9–28. http://doi.org/10.5130/AJCEB.v24i1/2.8906.

Bertot, J. C., P. T. Jaeger, and D. Hansen. 2012. "The impact of polices on government social media usage: Issues, challenges, and recommendations." *Gov. Inf. Q.* 29 (1): 30–40. https://doi.org/10.1016/j.giq.2011.04.004.

Biddulph, M. 2011. "Urban design, regeneration and the entrepreneurial city." *Prog. Plann.* 76 (2): 63–103. https://doi.org/10.1016/j.progress.2011.08.001.

Birjali, M., M. Kasri, and A. Beni-Hssane. 2021. "A comprehensive survey on sentiment analysis: Approaches, challenges and trends." *Knowledge-Based Syst.* 226 (Aug): 107134. https://doi.org/10.1016/j.knosys.2021.107134.

Blei, D. M., A. Y. Ng, and M. I. Jordan. 2003. "Latent dirichlet allocation." *J. Mach. Learn. Res.* 3 (4–5): 993–1022.

Das, R., and T. D. Singh. 2023. "Multimodal sentiment analysis: A survey of methods, trends, and challenges." *ACM Comput. Surv.* 55 (13S): 270. https://doi.org/10.1145/3586075.

Ezzeldin, M., and W. El-Dakhakhni. 2020. "Metaresearching structural engineering using text mining: Trend identifications and knowledge gap discoveries." *J. Struct. Eng.* 146 (5): 04020061. https://doi.org/10.1061/(ASCE)ST.1943-541X.0002523.

Flyvbjerg, B. 2012. "Why mass media matter to planning research: The case of megaprojects." *J. Plan. Educ. Res.* 32 (2): 169–181. https://doi.org/10.1177/0739456X12441950.

Gao, X., S. X. Zeng, R. C. Zeng, J. J. Shi, and R. Z. Song. 2022. "Multiple-stakeholders' game and decision-making behaviors in green management of megaprojects." *Comput. Ind. Eng.* 171 (Sep): 108392. https://doi.org/10.1016/j.cie.2022.108392.

Gao, Z., S. Q. Wang, and J. Gu. 2020. "Public participation in smart-city governance: A qualitative content analysis of public comments in urban China." *Sustainability* 12 (20): 8605. https://doi.org/10.3390/su12208605.

GMP (Gerkan, Marg und Partner). 2020. "Silk road international conference center." Accessed February 6, 2025. https://www.gmp.de/en/projects/9488/silk-road-international-conference-center.

Hemmatian, F., and M. K. Sohrabi. 2019. "A survey on classification techniques for opinion mining and sentiment analysis." *Artif. Intell. Rev.* 52 (3): 1495–1545. https://doi.org/10.1007/s10462-017-9599-6.

Hofmann, T. 2001. "Unsupervised learning by probabilistic latent semantic analysis." *Mach. Learn.* 42 (1): 177–196. https://doi.org/10.1023/A:1007617005950.

Jähnichen, P., F. Wenzel, and S. Mandt. 2018. Scalable generalized dynamic topic models." In *Proc., Int. Conf. on Artificial Intelligence and Statistics*, 1427–1435. Lanzarote, Spain: Proceedings of Machine Learning Research.

Jiang, H., P. Lin, and M. Qiang. 2016. "Public-opinion sentiment analysis for large hydro projects." *J. Constr. Eng. Manage.* 142 (2): 05015013. https://doi.org/10.1061/(ASCE)CO.1943-7862.0001039.

Jin, S. 2025. "Current status and trends of megaproject research: Bibliometric and text mining analysis." *Eng. Constr. Archit. Manage.* https://doi.org/10.1108/ECAM-04-2024-0495.

Koller, D., and N. Friedman. 2009. *Probabilistic graphical models: Principles and techniques*. Cambridge, MA: MIT Press.

Kumi, S., C. Snow, R. K. Lomotey, and R. Deters. 2024. "Uncovering concerns of citizens through machine learning and social network sentiment analysis." *IEEE Access* 12 (Jul): 94885–94913. https://doi.org/10.1109/ACCESS.2024.3426329.

Kundu, O., A. D. James, and J. Rigby. 2023. "Public opinion on megaprojects over time: Findings from four megaprojects in the UK." *Public Manage. Rev.* 25 (6): 1015–1038. https://doi.org/10.1080/14719037.2021.2003107.

Lee, C., J. W. Won, W. Jang, W. Jung, and Y. H. Kwak. 2017. "Social conflict management framework for project viability: Case studies from Korean megaprojects." *Int. J. Project Manage.* 35 (8): 1683–1696. https://doi.org/10.1016/j.ijproman.2017.07.011.

Li, H. Y., X. L. Zhang, S. T. Ng, M. Skitmore, and Y. H. Dong. 2018. "Social sustainability indicators of public construction megaprojects in China." *J. Urban Plann. Dev.* 144 (4): 04018034. https://doi.org/10.1061/(ASCE)UP.1943-5444.0000472.

Liang, Q., S. Ranganathan, K. Wang, and X. Deng. 2023. "JST-RR model: Joint modeling of ratings and reviews in sentiment-topic prediction." *Technometrics* 65 (1): 57–69. https://doi.org/10.1080/00401706.2022.2063187.

Lin, C., and Y. He. 2009. "Joint sentiment/topic model for sentiment analysis." In *Proc., 18th ACM Conf. on Information and Knowledge Management*, 375–384. New York: Association for Computing Machinery.

Lobo, S., and A. F. Abid. 2020. "The role of social media in intrastakeholder strategies to influence decision making in a UK infrastructure megaproject: Crossrail 2." *Project Manage. J.* 51 (1): 96–119. https://doi.org/10.1177/8756972819864456.

Luo, S. L., and S. Y. He. 2021. "Understanding gender difference in perceptions toward transit services across space and time: A social media mining approach." *Transp. Policy* 111 (Sep): 63–73. https://doi.org/10.1016/j.tranpol.2021.07.018.

Ma, L., and H. W. Fu. 2020. "Exploring the influence of project complexity on the mega construction project success: A qualitative comparative analysis (QCA) method." *Eng. Constr. Archit. Manage.* 27 (9): 2429–2449. https://doi.org/10.1108/ECAM-12-2019-0679.

Marcondes, F. S., A. Gala, R. Magalhães, F. Perez De Britto, D. Durães, and P. Novais. 2025. *Natural language analytics with generative large-language models: A practical approach with Ollama and open-source LLMs*, 23–35. Cham, Switzerland: Springer.

Mei, Q., X. Ling, M. Wondra, H. Su, and C. Zhai. 2007. Topic sentiment mixture: Modeling facets and opinions in weblogs." In *Proc., 16th Int. Conf. on World Wide Web*, 171–180. New York: Association for Computing Machinery.

Min, B. N., H. Ross, E. Sulem, A. P. Ben Veyseh, T. H. Nguyen, O. Sainz, E. Agirre, I. Heintz, and D. Roth. 2024. "Recent advances in natural language processing via large pre-trained language models: A survey." *ACM Comput. Surv.* 56 (2): 30. https://doi.org/10.1145/3605943.

Nik-Bakht, M., and T. E. El-Diraby. 2020. "Beyond chatter: Profiling community discussion networks in urban infrastructure projects." *J. Infrastruct. Syst.* 26 (3): 05020006. https://doi.org/10.1061/(ASCE)IS.1943-555X.0000555.

Nokkaew, M., et al. 2023. "Analyzing online public opinion on Thailand-China high-speed train and Laos-China railway mega-projects using advanced machine learning for sentiment analysis." *Social Network Anal. Min.* 14 (1): 15. https://doi.org/10.1007/s13278-023-01168-8.

Nokkaew, M., K. Nongpong, T. Yeophantong, P. Ploykitikoon, W. Arjharn, D. Phonak, A. Siritaratiwat, and C. Surawanitkun. 2024. "Hidden emotional trends on social media regarding the Thailand-China high-speed railway project: A deep learning approach with ChatGPT integration." *Social Network Anal. Min.* 14 (1): 175. https://doi.org/10.1007/s13278-024-01340-8.

Rodriguez-Melo, A., and S. A. Mansouri. 2011. "Stakeholder engagement: Defining strategic advantage for sustainable construction." *Bus. Strategy Environ.* 20 (8): 539–552. https://doi.org/10.1002/bse.715.

Sainati, T., G. Locatelli, N. Smith, N. Brookes, and G. Olver. 2020. "Types and functions of special purpose vehicles in infrastructure megaprojects." *Int. J. Project Manage.* 38 (5): 243–255. https://doi.org/10.1016/j.ijproman.2020.05.002.

Sengupta, A., S. Roy, and G. Ranjan. 2021. "LJST: A semi-supervised joint sentiment-topic model for short texts." *SN Comput. Sci.* 2 (4): 256. https://doi.org/10.1007/s42979-021-00649-x.

Soomro, S., et al. 2024. "How effective is Twitter (X) social media data for urban flood management?" *J. Hydrol.* 634 (May): 131129. https://doi.org/10.1016/j.jhydrol.2024.131129.

Tang, L., Y. Zhang, F. Dai, Y. Yoon, Y. Song, and R. S. Sharma. 2017. "Social media data analytics for the U.S. construction industry: Preliminary study on Twitter." *J. Manage. Eng.* 33 (6): 04017038. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000554.

TJAD [Tongji Architectural Design (Group)]. 2020. "Xi'an silk road international conference center." Accessed February 6, 2025. http://www.tjad.cn/project/686.

Tu, H., and S. Jin. 2024. "A group decision-making model for architectural programming in megaprojects." *Eng. Constr. Archit. Manage.* 31 (13): 342–368. https://doi.org/10.1108/ECAM-03-2024-0394.

Ural, E., and Y. C. Severcan. 2022. "Assessing user preferences regarding military site regeneration: The case of the Fourth Corps Command in Ankara." *Cities* 129 (Oct): 103807. https://doi.org/10.1016/j.cities.2022.103807.

Vijayvergia, A., and K. Kumar. 2021. "Selective shallow models strength integration for emotion detection using GloVe and LSTM." *Multimedia Tools Appl.* 80 (18): 28349–28363. https://doi.org/10.1007/s11042-021-10997-8.

Wang, C., D. Blei, and D. Heckerman. 2012. "Continuous time dynamic topic models." Preprint, submitted July 22, 2020. https://arxiv.org/abs/1206.3298.

Wang, D., and S. Li. 2025. "Social conflicts and their resolution paths in the commercialized renewal of old urban communities in China under the perspective of public value." *J. Urban Manage.* 14 (2): 402–417. https://doi.org/10.1016/j.jum.2024.11.007.

Wang, G., P. Wu, X. Y. Wu, H. J. Zhang, Q. Q. Guo, and Y. J. Cai. 2020. "Mapping global research on sustainability of megaproject management: A scientometric review." *J. Cleaner Prod.* 259 (Jun): 120831. https://doi.org/10.1016/j.jclepro.2020.120831.

Wang, L. X., J. F. Yuan, Y. D. Chen, X. Wan, and G. Y. Huang. 2024. "Intelligent construction benefits the public: Evidence from the opinion analysis on social media." *Eng. Constr. Archit. Manage.* https://doi.org/10.1108/ECAM-09-2024-1250.

Wang, M., R. Y. Ma, G. Q. Shen, and J. Xue. 2025. "How large language models empower the analysis of online public engagement for mega infrastructure projects: Cases in Hong Kong." *IEEE Trans. Eng. Manage.* 72 (Mar): 1262–1280. https://doi.org/10.1109/TEM.2025.3553595.

Wang, Y., Q. Han, B. de Vries, and J. Zuo. 2016. "How the public react to social impacts in construction projects? A structural equation modeling study." *Int. J. Project Manage.* 34 (8): 1433–1448. https://doi.org/10.1016/j.ijproman.2016.07.008.

Wu, L. F., G. S. Jia, and N. Mackhaphonh. 2019. "Case study on improving the effectiveness of public participation in public infrastructure megaprojects." *J. Constr. Eng. Manage.* 145 (4): 05019003. https://doi.org/10.1061/(ASCE)CO.1943-7862.0001623.

Xue, J., G. Shen, Y. Li, J. Wang, and I. Zafar. 2020. "Dynamic stakeholder-associated topic modeling on public concerns in megainfrastructure projects: Case of Hong Kong–Zhuhai–Macao bridge." *J. Manage. Eng.* 36 (6): 04020078. https://doi.org/10.1061/(ASCE)ME.1943-5479.0000845.

Xue, J., G. Q. Shen, Y. Li, S. Han, and X. Chu. 2021. "Dynamic analysis on public concerns in Hong Kong-Zhuhai-Macao bridge: Integrated topic and sentiment modeling approach." *J. Constr. Eng. Manage.* 147 (6): 04021049. https://doi.org/10.1061/(ASCE)CO.1943-7862.0002066.

Yan, Z., X. Jing, and W. Pedrycz. 2017. "Fusing and mining opinions for reputation generation." *Inf. Fusion* 36 (Jul): 172–184. https://doi.org/10.1016/j.inffus.2016.11.011.

Yang, J., X. M. Zou, W. Zhang, and H. Y. Han. 2021. "Microblog sentiment analysis via embedding social contexts into an attentive LSTM." *Eng. Appl. Artif. Intell.* 97 (Jan): 104048. https://doi.org/10.1016/j.engappai.2020.104048.

Yao, L., Y. Zhang, B. Wei, L. Li, F. Wu, P. Zhang, and Y. Bian. 2016. "Concept over time: The combination of probabilistic topic model with Wikipedia knowledge." *Expert Syst. Appl.* 60 (Oct): 27–38. https://doi.org/10.1016/j.eswa.2016.04.014.

Yin, L., M. Han, and X. Y. Nie. 2024. "Unlocking blended emotions and underlying drivers: A deep dive into COVID-19 vaccination insights on twitter across digital and physical realms in New York, using ChatGPT." *Urban Sci.* 8 (4): 222. https://doi.org/10.3390/urbansci8040222.

Yu, J. B., J. H. Zhao, L. Miranda-Moreno, and M. Korp. 2025. "Modular AI agents for transportation surveys and interviews: Advancing engagement, transparency, and cost efficiency." *Commun. Transp. Res.* 5 (Dec): 100172. https://doi.org/10.1016/j.commtr.2025.100172.

Yu, S. J. 2022. "Urban collage: How can affective dynamics shape waterfront landscapes?" *Landscape Res.* 47 (1): 68–86. https://doi.org/10.1080/01426397.2021.1978410.

Zander, K. K., R. Nepal, and S. T. Garnett. 2024. "Assessing good governance principles of renewable energy megaprojects." *J. Cleaner Prod.* 477 (Oct): 143848. https://doi.org/10.1016/j.jclepro.2024.143848.

Zhong, B., J. Tian, X. Pan, and L. Shen. 2024. "An online housing reputation assessment framework based on text mining and visualization technologies." *J. Constr. Eng. Manage.* 150 (8): 04024076. https://doi.org/10.1061/JCEMD4.COENG-14669.